Module 6 | Project Data Cleaning

AI Workforce Investment Planner - Vision A

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Prepared October 4, 2026 with Codex assistance. Individual evidence report for team reconciliation; this is not a claim that teammates have reviewed or reconciled these results.

Question: Which California OEWS area-occupation combinations warrant further review using published employment, median annual wages, and state-level observed Claude usage? These inputs support descriptive screening, not causal claims or an investment ranking.

Verified output	Result
California area-occupation rows	11,847
Areas / distinct occupations	28 / 783
Selected Anthropic occupation-metric rows	149,421
Missing employment / median annual wage	215 / 169
Rows with no published matching CA AI share	5,387
Fresh notebook reproduction	Identical cleaned-file hashes

Read before using the result

The joined AI share is repeated state-level context, not a local adoption estimate. The O*NET-to-SOC prefix mapping is provisional and must be verified before decision use. Missing AI cells are not zero. No priority score is calculated. Team reconciliation and the final Gradescope upload remain separate human steps.

Evidence location: outputs/evidence/. Full cleaned tables are regenerated by clean_data.py and included in the local ZIP; GitHub contains the 50-row sample and full-data evidence.

Checklist | Inspection and cleaning

1. State vision and question - Done

Vision A confirmed by the student. Compare published employment and annual median wages across California OEWS areas; use CA Claude occupation shares as separate context. See cleaning_plan.md. Business prioritization weights are not inferred.

2. Define one row - Done

BLS: one area x detailed occupation, unique AREA + OCC_CODE after filtering. AEI: one product x period x geography x category x hierarchy x metric x node. Signals: one published grouping-key combination; key definitions are in full_data_checks.json. A repeated occupation identifier across areas/periods/metrics is valid.

3. Preserve source - Done

Original files are opened read-only. source_manifest.json records SHA-256 and size for all seven original input files. BLS release: May 2025; AEI release: June 26, 2026; Signals package: local v2.0 package. Original retrieval dates are unknown; inspection was October 4, 2026. Hashes are checked again after execution.

4. Inspect columns and types - Done

All selected original source rows were inspected. Strings preserve leading zeros, including nonmetro area IDs and NAICS 000000. Dates are checked against YYYY-MM-DD; numeric conversions reject unexpected tokens. All unexpected conversion failures are zero. BLS blank/*/**/# markers are recognized missing values, not failed numeric conversions. Raw markers and values remain alongside typed fields. See per-source conversion_failures, numeric_markers_full, missing_before and missing_after.

5. Standardize text - Done

Boundary whitespace is trimmed without changing capitalization or punctuation. 0 source cells changed. No boundary-whitespace changes were needed; unchanged identifiers include 0600003 -> 0600003 and 000000 -> 000000. Source labels are retained in raw columns for traceability. See text_changes/text_examples in full_data_checks.json.

6. Investigate duplicates - Done

Exact duplicate excess and full-key repetition counts are recorded separately for every source. All are zero in this run. Node IDs repeat legitimately across periods, metrics and geographies. No rows were deduplicated; unexpected logical-key collisions would stop execution.

7. Handle missing data - Done

Final BLS employment has 215 missing values; annual median wage has 169. Recognized suppression/top-code markers become numeric missing values, while raw tokens remain. No imputation or missing-driven row drops occur. 5,387 unmatched AI rows stay missing. Per-field before/after counts and marker counts appear in full_data_checks.json. Filling with zero would understate unknown values.

Checklist | Validation

8. Check suspicious values - Done

All AEI values and dates were parsed and checked; percent bounds are 0-100 and autonomy bounds 1-5. Signals shares use 0-1. BLS selected numeric fields are checked for negative values. Range flags found: 0. Top-coded wages remain missing rather than using a fabricated exact wage. No outlier trimming is applied; unusual values are not automatically errors.

9. Verify merges - Unresolved

The mechanical join passes many-to-one validation: 11,847 BLS rows become 11,847 joined rows, with no row multiplication. 460 CA leaf cells aggregate to 401 prefixes. 5,387 BLS rows (415 codes) are unmatched; 33 AI prefixes have no selected BLS occupation. Exported unmatched tables document both sides. The prefix mapping is provisional, not a verified taxonomy crosswalk; this prevents final investment conclusions.

10. Reconcile changes - Done

BLS originals contain 48,689 nonmetro and 150,023 metro rows. Declared geography/occupation filters retain 973 + 10,874 = 11,847; all exclusions are accounted for per workbook. AEI originals contain 2,128,278 rows; the documented occupation/geography scope retains 149,421. Four Signals files retain all 8,328 rows. Published BLS employment totals 17,277,510 before and after the join; this is a partial published detailed-occupation sum, not a complete state employment estimate. CA leaf share total 97.82% is preserved by aggregation.

11. Check calculations - Done

Employment sum uses 11,632 nonmissing rows; wage summaries use 11,678. area_metric_support.csv reports group-specific denominators. Missing values are excluded from sums/means; group keys are retained with dropna=False. The average of occupation median wages is explicitly unweighted and is not an employee-level median. State AI shares are never summed over repeated areas. No cross-product/time averaging is used.

12. Verify five records - Done

Independent validation checks every selected BLS row and its 19 numeric columns against the original XLSX files. It also independently aggregates original CA AEI rows using Decimal and verifies every joined row. Five difficult examples follow and are saved in independent_validation.json.

13. Test reproducibility - Done

The submitted notebook ran in a fresh kernel and reproduced all 9 cleaned files with identical SHA-256 hashes. The independent validator passed. See reproducibility.json, output_hashes.json and the executed cleaning.ipynb. Exact commands and environment dependencies are in README.md.

14. Remaining limitations - Done

Documented limits: provisional taxonomy alignment; incomplete published AI cells; AI-use share differs from exposure/adoption; state AI values do not measure metro usage; BLS and AI vintages differ; published employment excludes suppressed cells; no causal outcome or ROI; team reconciliation pending. These limits constrain interpretation even when transformations are correct.

Cleaning decision log

Rule	Decision	Reason / effect
R1	Source preservation	Hash original files; no source edits. Retrieval date remains unknown.
R2	Scope	CA detailed OEWS areas plus declared AEI occupation/geography scope and four Signals context tables.
R3	Text and identifiers	Trim only boundary whitespace; retain meaningful labels and leading zeros.
R4	Missing values	Convert recognized BLS markers to missing with raw markers retained; no zero fill.
R5	Duplicates	Check exact rows and full keys separately; stop on collisions; remove none.
R6	Geography and units	Keep product, date, geography and units visible; API remains global context.
R7	Join	Aggregate same-period published CA leaf shares by SOC prefix; left join many-to-one; mapping provisional.
R8	Unusual values	Check appropriate ranges; flag rather than drop or cap.
R9	Validation/sample	Full-data assertions, independent source checks, deterministic difficult-case sample and fresh rerun.

The full pre-implementation plan and revision log are in cleaning_plan.md. No analytical rule was revised after observing results; implementation-only fixes are recorded there. These are Codex-assisted proposed choices, not evidence of team approval.

Five original-to-cleaned checks

Values below are employment; annual median wage (USD); California Claude usage (%). Numeric display is rounded for readability; exact source/output values and comparisons are preserved in independent_validation.json.

1. 0600003 / 27-3041 - Editors

Source: oesm25ma/BOS_M2025_dl.xlsx, data record 3353 (worksheet row = record + 1).

Original employment / wage: ** / 71300

Expected: missing; 71300; 2.49 | Actual: missing; 71300; 2.49 | MATCH

Unpublished employment stays missing, not zero.

2. 0600006 / 29-1211 - Anesthesiologists

Source: oesm25ma/BOS_M2025_dl.xlsx, data record 3699 (worksheet row = record + 1).

Original employment / wage: ** / #

Expected: missing; missing; missing | Actual: missing; missing; missing | MATCH

Unpublished employment stays missing, not zero. Top-coded wage stays missing; its exact value is unknown. Unpublished/unmatched AI cell stays missing, not zero.

3. 0600003 / 19-4042 - Environmental Science and Protection Technicians, Including Health

Source: oesm25ma/BOS_M2025_dl.xlsx, data record 3304 (worksheet row = record + 1).

Original employment / wage: 90 / *

Expected: 90; missing; missing | Actual: 90; missing; missing | MATCH

Unavailable wage stays missing, not zero. Unpublished/unmatched AI cell stays missing, not zero.

4. 0600003 / 11-9021 - Construction Managers

Source: oesm25ma/BOS_M2025_dl.xlsx, data record 3239 (worksheet row = record + 1).

Original employment / wage: 190 / 117310

Expected: 190; 117310; missing | Actual: 190; 117310; missing | MATCH

Unpublished/unmatched AI cell stays missing, not zero.

5. 0600003 / 11-3031 - Financial Managers

Source: oesm25ma/BOS_M2025_dl.xlsx, data record 3235 (worksheet row = record + 1).

Original employment / wage: 380 / 130990

Expected: 380; 130990; 0.41 | Actual: 380; 130990; 0.41 | MATCH

Sum only published leaf-code shares within one CA period; mapping remains provisional.

Codex verification response

Verification performed by Codex on October 4, 2026. I executed the cleaning script, independently checked source-to-output transformations, and executed the notebook in a fresh kernel. Every declared source was inspected in full; the sample was not used as a substitute for full-data checks. All selected BLS rows were independently compared with the original workbook values, and every joined AI value was independently recomputed from the original AEI file. Cleaned output hashes matched on the notebook rerun.

What this verifies: the implemented rules reproduce the submitted outputs; identifiers, suppression markers, numeric values, row counts, join cardinality, and selected totals are consistent with original sources. What it does not verify: that the provisional occupation-prefix mapping is an official crosswalk, that a state usage share measures regional adoption, that cleaning choices have received student/team approval, or that these inputs establish pilot effectiveness. Those decisions remain with the analyst/team.

Team reconciliation / submission status

Pending: compare each member's independent cleaning results, reconcile scope and mapping choices, approve the final group checklist, and upload the group PDF/ZIP to Gradescope. This report supplies individual evidence for that discussion. It does not assert reconciliation or Gradescope submission.

A preliminary comparison with Qiyue's published branch is in team_reconciliation.md. Both workflows report 11,847 BLS rows; the AI scopes and integrated row units differ. Matching BLS counts alone does not establish matching values or team agreement.

Source and access references

Anthropic June 2026 schema and publication thresholds: https://huggingface.co/datasets/Anthropic/EconomicIndex/blob/main/release_2026_06_26/data_documentation.md

BLS May 2025 OEWS tables: https://www.bls.gov/oes/2025/may/oes_2025_m.htm

BLS original regional archive: https://www.bls.gov/oes/special.requests/oesm25ma.zip

OpenAI Signals package and methodology: https://openai.com/signals/data-download/

Anthropic Economic Index data attribution: Massenkoff et al., Anthropic Economic Index report: Cadences (June 26, 2026), CC-BY. OpenAI Signals: OpenAI, v2.0 local data package; consult its included README.pdf and download-page terms. BLS: U.S. Bureau of Labor Statistics, May 2025 OEWS. Preserve attribution when redistributing samples.